

Technology Stack

Date	27 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend Client-Side	React.js, Vite, Tailwind CSS, React Router DOM	Chosen for responsive UI design, fast rendering, component reusability, modern SPA architecture, and smooth user experience.
2	Backend Server-Side	FastAPI, Python	Chosen for fast API processing, high performance, asynchronous support, easy AI integration, and scalable backend development.

3	Database / Data Storage	SQLite, SQLAlchemy ORM, PostgreSQL-ready Architecture	Selected for lightweight development, strong relational mapping, structured financial data management, and future production scalability.
4	Cloud / Hosting / Deployment	Vercel, Render	Chosen for cost-effective hosting, easy deployment, GitHub integration, scalability, and reliable frontend-backend cloud hosting.
5	Version Control & CI/CD	GitHub, Git	Used for seamless team collaboration, version tracking, source code management, automated deployment integration, and project maintenance.
6	Third-Party APIs / Other Tools	Google Gemini API, SMTP Email Service	Gemini API enables AI-powered settlement prediction and negotiation letter generation, while SMTP enables email verification, password reset, and notification services.

